



THE TRANSFORMATIVE POTENTIAL OF SPATIAL COMPUTING FOR DUBAI

MARKET STUDY

MAY 10, 2025

PREPARED BY FUTURISTS HUB

Spatial Computing: The Next Technological Frontier

- **Definition & Scope**

Technology that **merges digital information with physical world**, creating immersive and interactive experiences through AR, VR, and MR. Enables machines to understand and interact with real-world geometry and context.

- **Touchable AI: A Paradigm Shift**

Spatial computing enables **Touchable AI** - the evolution where artificial intelligence becomes **physically interactive and tangible** in our environment rather than just existing behind screens. This transforms human-AI interaction through **natural gestures, voice, and movement** in shared physical spaces.

- **Global Market Trajectory**

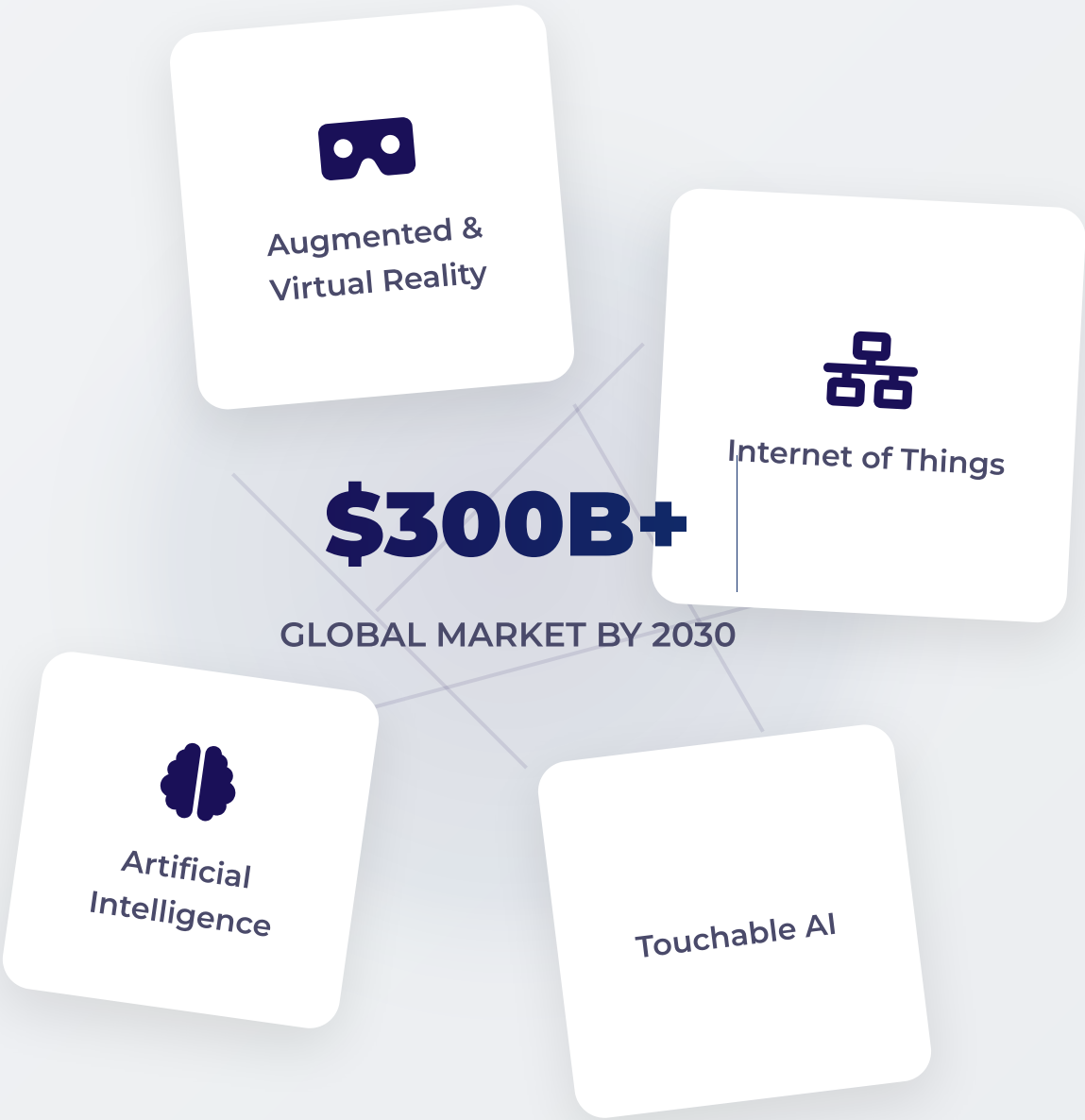
Projected to reach **\$300+ billion by 2030**, growing at a CAGR of 35%. Driven by advancements in AR/VR hardware, AI integration, and increasing enterprise adoption across healthcare, manufacturing, retail, and entertainment.

- **Dubai's Strategic Position**

Well-positioned to become a **leading hub for spatial computing** with its ambitious Dubai Metaverse Strategy, existing smart city infrastructure, and forward-thinking regulatory environment. Aligns with Dubai's vision for technological leadership.

- **Strategic Opportunities for Dubai**

Potential to **pioneer regional innovation ecosystem** through specialized infrastructure development, technology partnerships, talent attraction, and regulatory frameworks that position Dubai as the MENA region's center of excellence for spatial computing technologies.



What is Spatial Computing?

A **paradigm shift** in human-computer interaction that **merges digital information with the physical world**, creating immersive and interactive experiences. It enables machines to understand and interact with the geometry and context of the real world, and for humans to interact with digital information in a more **natural and intuitive** way.

Touchable AI: The Future of Interaction

Spatial computing represents the evolution toward **Touchable AI** — where artificial intelligence becomes physically interactive and tangible in our environment. Unlike traditional AI that exists behind screens, **Touchable AI** can be manipulated physically, responds to spatial context, and integrates seamlessly with our physical surroundings. This allows for **intuitive interactions with digital intelligence** through natural gestures, voice, and movement, bridging the gap between humans and AI systems through shared physical spaces.



Spatial Awareness

Systems that can perceive, map, and understand physical environments and spatial relationships in real-time, creating a digital representation of the real world.



3D Model Integration

Ability to create and manipulate accurate 3D models and digital twins, enabling simulation, analysis, and visualization of physical assets.



Physical Interaction

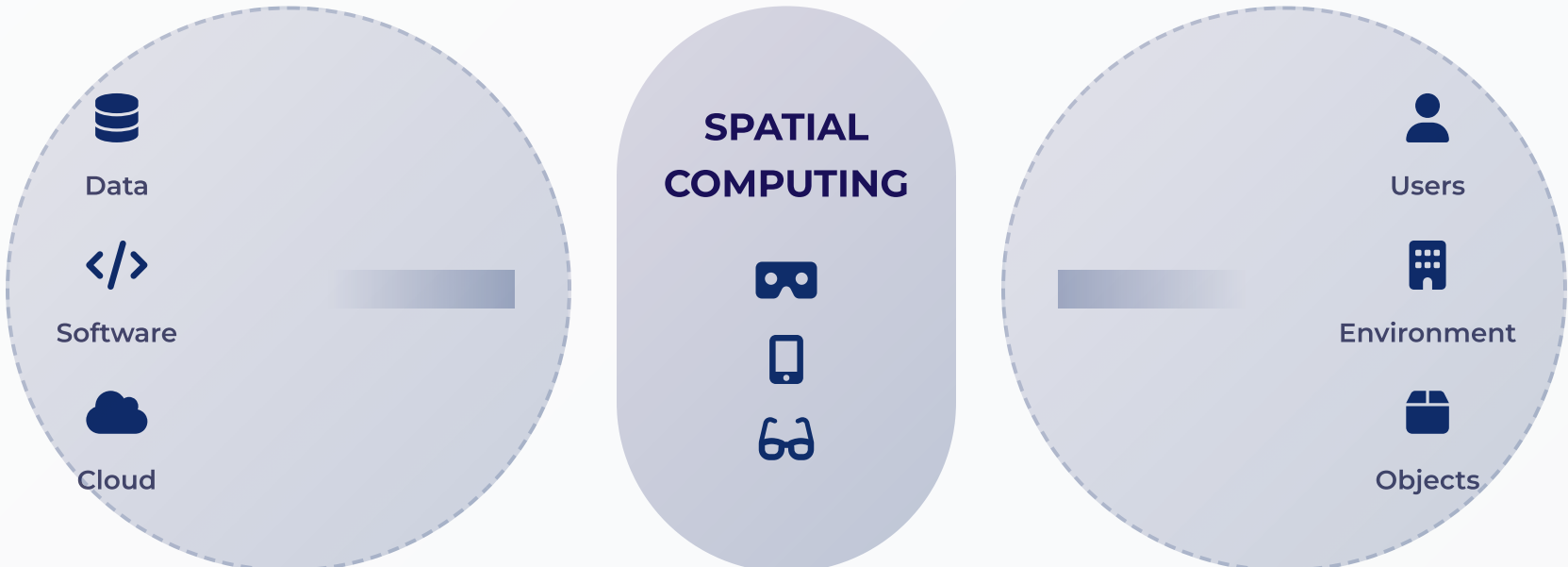
Natural interfaces allowing users to interact with digital content using physical gestures, voice commands, and movement within real-world space.



AI-Driven Decision Making

Artificial intelligence that processes spatial data, recognizes objects, and makes real-time decisions to create contextually relevant experiences.

How Spatial Computing Works



The Technologies Powering Spatial Computing

Touchable AI



The Evolution of Artificial Intelligence in Spatial Computing

Touchable AI represents the next frontier in human-machine interaction, transforming artificial intelligence from screen-based interfaces to **physically interactive and tangible** intelligence within our spatial environment. Unlike traditional AI that exists behind screens, Touchable AI can be **manipulated with natural gestures**, responds to spatial context, and integrates seamlessly with our physical surroundings. This evolution is central to spatial computing's value proposition, creating intuitive environments where digital intelligence becomes a **touchable, physical extension** of human capability rather than a separate digital entity.



AR/VR/MR

The visual and interactive interface for spatial computing. **AR overlays digital information** onto the real world, **VR creates immersive digital environments**, and **MR blends real and virtual elements** that can interact in real time.



AI & Machine Learning

Crucial for **processing sensory data**, enabling **environment understanding**, object recognition, natural language processing, and **personalizing user experiences** within spatial contexts.



IoT & Sensor Networks

IoT devices and sensors **collect real-time data** about physical conditions, object locations, and user activities, enabling spatial computing systems to **react dynamically** to the physical world.



3D Modeling & Digital Twins

Creates **accurate 3D models** of objects, buildings, and cities. Digital twins (virtual replicas of physical assets) allow for **simulation, analysis, and remote management** of real-world systems.



Cloud & Edge Computing

Provides the **computational power** for processing spatial data. Edge computing brings computation closer to data sources, **reducing latency** and improving responsiveness for seamless experiences.



High-Speed Connectivity

5G and future 6G networks provide the reliable, high-bandwidth, low-latency communication essential for transmitting large volumes of spatial data, especially for **mobile applications**.

Strategic Significance of Spatial Computing



Enhancing Human Capabilities

Provides intuitive access to information and digital tools within the physical context, augmenting human intelligence, creativity, and productivity across professional and personal domains.



Transforming Industries

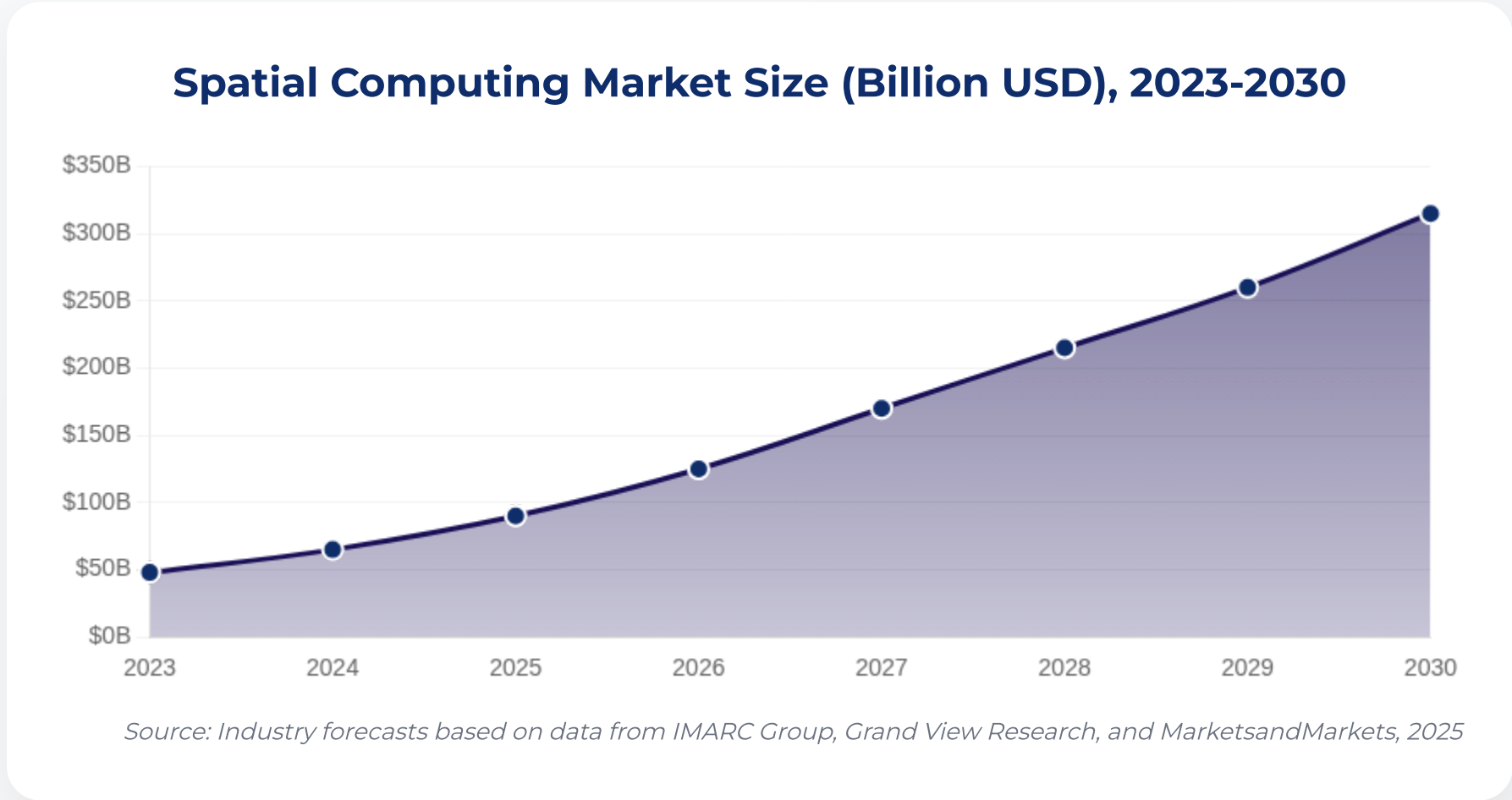
Revolutionizes sectors from manufacturing and healthcare to retail and entertainment, offering new ways to design, operate, and experience products and services.



Creating Economic Value

Drives significant economic growth by creating new markets, businesses, and job opportunities, while increasing efficiency and enabling innovative business models.

Global Market Opportunities & Growth Projections



Key Market Drivers



AR/VR Hardware Advancements

Lighter, more powerful devices with improved field of view, resolution, and battery life making spatial experiences more accessible and comfortable.



AI Integration

Machine learning and computer vision advances enabling more accurate environmental mapping, object recognition, and context-aware experiences.



Enterprise Adoption

Organizations increasingly implementing spatial computing for training, remote collaboration, design visualization, and operational efficiency.



User Experience Improvements

More intuitive interfaces, natural interaction methods, and compelling content driving user engagement and mainstream adoption.

High-Growth Industry Segments



Healthcare

CAGR 42%



Manufacturing

CAGR 38%



Retail

CAGR 35%



Education

CAGR 39%



Entertainment

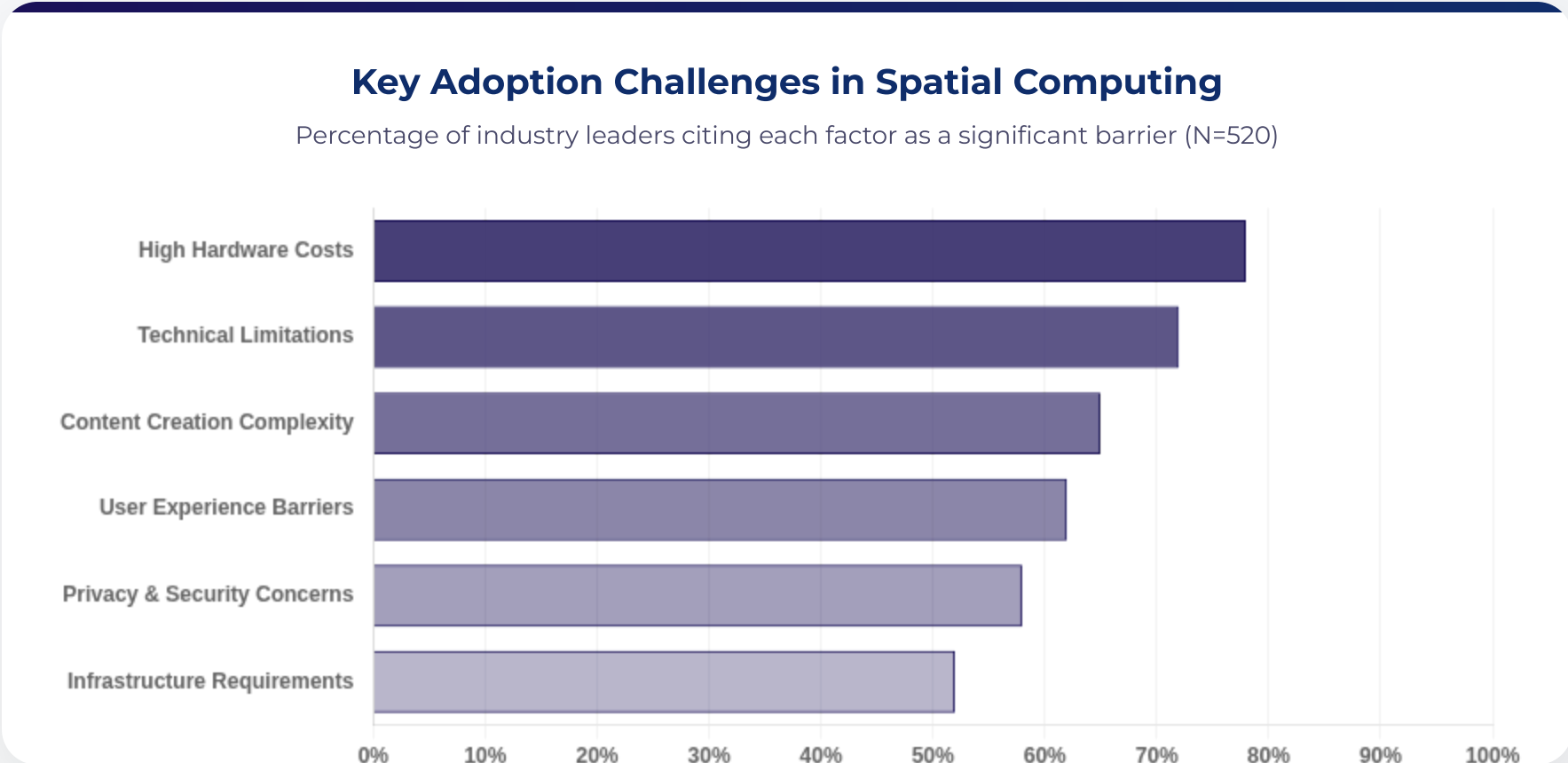
CAGR 40%

Regional Market Insights

North America	36% CAGR	MARKET LEADER
Asia Pacific	41% CAGR	FASTEST GROWING
Europe	33% CAGR	ESTABLISHED
Middle East & Gulf	38% CAGR	EMERGING LEADER
Latin America	31% CAGR	EMERGING

Navigating Barriers to Adoption & Implementation

Understanding and addressing these challenges is critical for the successful implementation of spatial computing technologies



\$

78% Critical

High Hardware Costs

Premium headsets and spatial devices cost \$2,000-\$3,500 per unit, limiting widespread adoption in consumer and enterprise markets.

58% price reduction expected by 2028 through component miniaturization

🔧

72% Critical

Technical Limitations

Current hardware faces constraints in field of view (52-90° vs. human 210°), resolution, processing power, and battery life (2-3 hours).

4x improvement in computing power-to-weight ratio expected in next generation

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65% Critical

Content Creation Complexity

Development costs 3-5x higher than traditional apps, with specialized skills required and fragmented development tools.

73% faster development with no-code/low-code spatial platforms

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58% Critical

Privacy & Security

Spatial devices capture extensive environmental data, with 62% of consumers expressing privacy concerns. Security frameworks still developing.

85% of data processing shifting to on-device computation by 2028

Dubai's Strategic Advantages

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World-Class Digital Infrastructure

97.8% 5G coverage and advanced edge computing nodes provide ideal foundation for data-intensive spatial applications.

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Agile Regulatory Environment

Forward-thinking regulatory approach, including Metaverse Self-Governance principles for privacy and security standards.

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Talent Development Ecosystem

Partnerships with 14 global technology universities to address talent gaps in spatial computing.

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Strategic Investment Capacity

\$50M+ allocated to metaverse technologies to subsidize adoption and specialized infrastructure.

Emerging Trends Addressing These Challenges

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AI-Generated Content

Generative AI reducing content creation costs by 60-70% through automated 3D asset creation and personalization.

2025-2026

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Lightweight Devices

Smaller, lighter devices (under 120g) similar to conventional eyewear with 4+ hours of battery life.

2026-2027

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Open Standards

Industry consortiums developing interoperable frameworks with 85% cross-platform compatibility targets.

2025-2028

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XaaS Models

Spatial Computing as a Service (\$15-45/user/month) reducing upfront investment by 70%.

2026-2027

Dubai's Vision: Pioneering a Phygital Future

Smart City Foundation

Smart Dubai 2021

Comprehensive digital infrastructure with IoT networks, 5G connectivity, and city-wide sensors creating the ideal foundation for spatial computing applications.

Digital Twin City Model

Detailed 3D mapping of Dubai's physical infrastructure enabling sophisticated spatial computing applications for urban planning and services.

Competitive Advantages

Tech-Forward Leadership

Government-led innovation strategy with dedicated funding for emerging technologies and commitment to digital transformation at the highest levels.

Innovation-Friendly Regulation

Responsive regulatory framework including the Metaverse Self-Governance Framework, virtual asset regulations, and regulatory sandboxes for testing.

Strategic Location

Global connectivity and position as regional tech hub enabling rapid adoption and deployment of spatial computing solutions across MENA region.

Talent & Capital Hub

Access to international tech talent, significant investment capital, and technology accelerators focused on metaverse and spatial computing innovation.

Current Implementations

Museum of the Future

Pioneering spatial computing exhibitions and immersive experiences showcasing the potential of blended physical-digital environments.

Virtual Property Tours

Real estate developers utilizing spatial computing for immersive property previews, enhancing global investment opportunities.

Digital Tourism Experiences

Augmented reality historical tours and spatial computing enhanced visitor experiences at major attractions and retail destinations.

Dubai Metaverse Strategy

Economic Impact

Aims to contribute \$4 billion to Dubai's economy by 2030, positioning Dubai among the top 10 metaverse economies globally and a leading spatial computing innovation hub.

Job Creation

Expected to support 42,000 virtual jobs by 2030, developing a robust talent pipeline for spatial computing and metaverse skills through strategic education initiatives.

Infrastructure Development

Investing in cutting-edge infrastructure to enable spatial computing applications across government services, tourism, education, retail, remote work, and healthcare.



Industry-Specific Opportunities & Applications



Tourism & Hospitality

- **AR-enhanced tourist experiences** overlaying historical information, directions, and personalized recommendations
- **Virtual previews** of attractions, hotels, and experiences before booking
- **Immersive cultural presentations** recreating historical Dubai and cultural traditions

Projected Growth: 45% CAGR (2025-2030)



Real Estate & Construction

- **Virtual property tours** with interactive customization for off-plan properties
- **AR-assisted construction** displaying BIM models on-site for improved accuracy
- **Digital twins** for property management, maintenance, and sustainability optimization

Projected Growth: 42% CAGR (2025-2030)



Retail & E-commerce

- **Virtual try-on solutions** for fashion, jewelry, and cosmetics
- **Immersive showrooms** for luxury goods and automotive displays
- **Personalized AR shopping assistants** providing product information and recommendations

Projected Growth: 38% CAGR (2025-2030)



Education & Training

- **Immersive learning environments** for STEM education and virtual field trips
- **VR-based training simulators** for medical, technical, and vocational education
- **Collaborative learning spaces** connecting international students and educators

Projected Growth: 40% CAGR (2025-2030)



Healthcare

- **Surgical planning and training** using patient-specific 3D models
- **Remote consultations** with spatial presence and holographic visualization
- **AR-guided procedures** with real-time data overlay for medical professionals

Projected Growth: 44% CAGR (2025-2030)

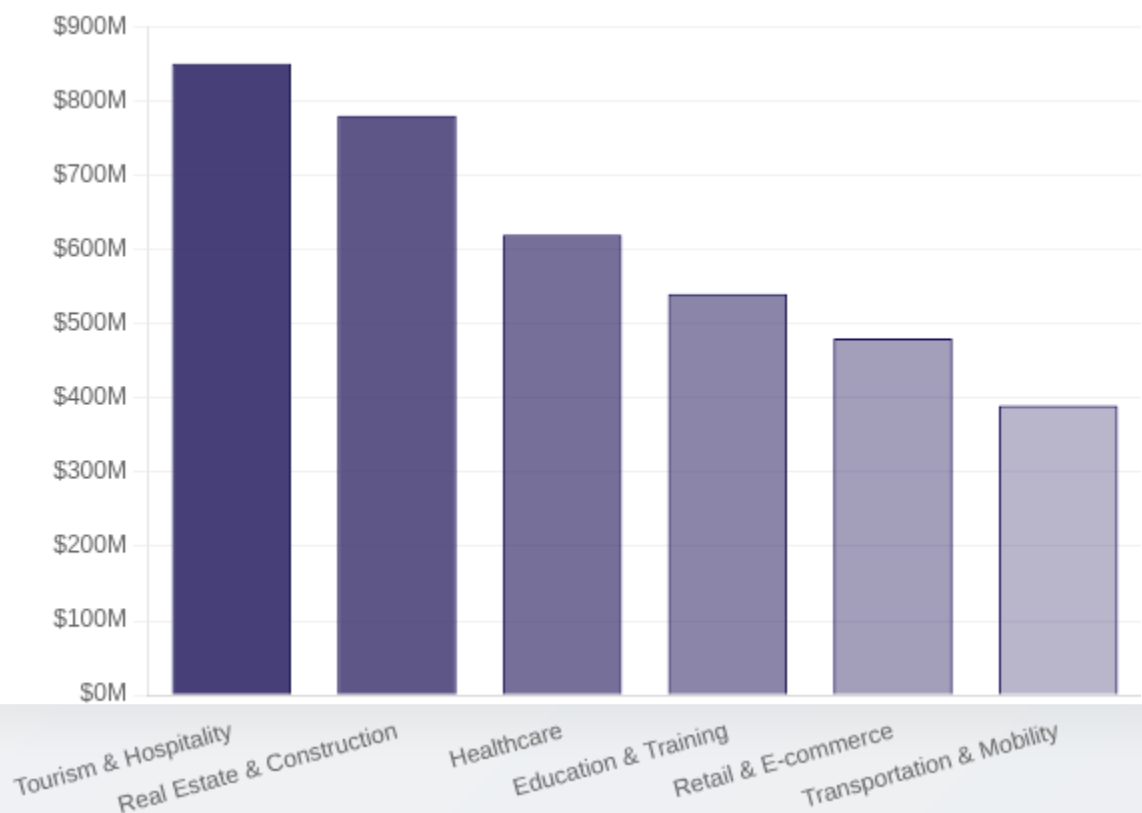


Transportation & Mobility

- **AR navigation systems** for pedestrians and autonomous vehicles
- **Spatial traffic management** with real-time digital twin of transport network
- **Enhanced passenger experiences** in metro stations and airports

Projected Growth: 36% CAGR (2025-2030)

Dubai Spatial Computing Market Potential by Sector (2030)



Strategic Partnerships & Innovation Ecosystem



Academic Partnerships

Collaborations with UAE and international universities for research, curriculum development, and talent pipeline creation in spatial computing.



Technology Accelerators

Dedicated spatial computing incubators providing mentorship, funding, and technical resources for startups developing innovative solutions.



Public-Private Partnerships

Government collaboration with private sector to develop use cases, pilot programs, and large-scale implementations across key sectors.



Global Tech Alliances

Strategic relationships with global technology leaders to bring cutting-edge spatial computing expertise and technology to Dubai.

Key Players Shaping the Spatial Computing Ecosystem

The market features diverse competitors across hardware, software platforms, and infrastructure



**Apple**
PREMIUM HARDWARE

Focus on high-end experience, tight hardware-software integration, and premium positioning with quality over mass adoption strategy.

Key Products:

- Vision Pro (\$3,499)
- Market Share: ~8%

**Meta Platforms**
METAVERSE PLATFORM

Mass market adoption through affordable hardware, subsidized by future ecosystem revenue and advertising potential.

Key Products:

- Quest Headsets (\$299-499)
- VR Market Share: ~65%

**Microsoft**
ENTERPRISE MR

Enterprise-focused with emphasis on industrial applications, remote assistance, and integration with existing Microsoft ecosystem.

Key Products:

- HoloLens & Mesh (\$3,500+)
- Enterprise Share: ~12%

**NVIDIA**
INFRASTRUCTURE

Enabling infrastructure through GPUs and development platforms, powering the underlying technology stack for spatial computing.

Key Products:

- Omniverse Platform
- GPU Market Share: 90%+

**Google**
MOBILE AR

Currently focused on smartphone-based AR through ARCore, with potential for future AR glasses in development with Samsung.

Key Products:

- ARCore Platform
- Mobile AR Share: ~80%

Specialized Technology Enablers

**Qualcomm**
XR Chipsets

**Unity/Epic**
3D Engines

**PTC**
Industrial AR

**OpenAI**
Generative AI

**Niantic**
Location AR

Transforming Industries Through Spatial Computing



Healthcare

42% CAGR through 2030

Surgical Planning & Navigation

3D visualization of patient anatomy with AR overlays during procedures, improving precision by 23%



Medical Training

VR simulations for risk-free procedural practice with unlimited repetition and performance tracking



Remote Consultation

Spatial telepresence allowing specialists to virtually assist during examinations

Result: 38% faster surgical planning

Leading medical centers report substantial improvements using AR-guided navigation for neurosurgery and orthopedics



Education

39% CAGR through 2030



Immersive Learning

Interactive 3D models for STEM subjects, transforming abstract concepts into tangible experiences



Collaborative Spaces

Shared virtual environments for geographically dispersed students to interact with content together



Virtual Field Trips

Access to otherwise inaccessible locations from ocean floors to distant historical sites

Result: 75% better information retention

Institutions report significantly higher engagement rates compared to traditional learning methods



Manufacturing

38% CAGR through 2030



AR-Guided Assembly

Visual instructions overlaid directly on equipment, reducing errors by 90% and improving efficiency



Digital Twins

Virtual replicas for simulation, optimization, and predictive maintenance of physical systems



Remote Expert Support

AR-enabled assistance where experts guide technicians with visual annotations

Result: 25% reduction in production time

Automotive manufacturers achieve 45% fewer quality issues with 6-8 month ROI on AR implementation



Retail & E-commerce

35% CAGR through 2030



Virtual Try-On

AR visualization of products on customers or in their spaces, reducing returns by up to 40%



Immersive Shopping

Virtual stores where shoppers browse 3D products, blending online convenience with physical experiences



In-Store AR Navigation

Personalized guidance, product information, and promotions based on shopper location

Result: 65% higher engagement

Luxury retailers report 32% increase in conversion rates with virtual try-on for accessories

Additional Industry Applications



Entertainment & Media



Architecture & Construction



Transportation



Urban Planning



Defense & Security

Enabling Dubai's Leadership in Spatial Computing

Touchable AI Initiative

Position Dubai as the Global Leader in Interactive AI

Dubai should pioneer **Touchable AI** - a revolutionary approach where artificial intelligence becomes physically interactive and seamlessly integrated into the urban landscape. This creates a **distinctive advantage** other tech hubs cannot easily replicate, building on Dubai's smart city infrastructure to enable AI interactions through natural gestures in public spaces, transportation, retail, and tourism.

**Urban Integration**

AI systems throughout Dubai's infrastructure that respond to physical interaction

**Citizen Experience**

Intuitive public services with natural gesture, voice and movement interaction

**Global Differentiation**

Establish Dubai as the world's first "Touchable AI City"

**Economic**

Investment Fund

Establish a **\$500M fund** for venture capital, grants, and matching investments for spatial computing innovations

Tax Incentives

Offer **tax exemptions** for R&D, equipment imports, and reduced registration fees

Government Procurement

Prioritize **spatial computing solutions** in public services to provide market validation

**Policy**

Regulatory Framework

Develop regulations for privacy, data ownership, and content standards building on **Metaverse Self-Governance**

Innovation Licensing

Create **specialized categories** with streamlined processes and regulatory sandboxes

Standards Leadership

Lead **technical standards** development regionally and participate in global standards bodies

**Infrastructure**

Advanced Connectivity

Accelerate **5G/6G deployment** with ultra-low latency and strategic edge computing nodes to power spatial applications

City-Scale Digital Twin

Invest in a **comprehensive digital twin** of Dubai with precision 3D mapping and open APIs for developers

Innovation Hubs

Establish **dedicated centers** with specialized equipment and testing facilities for spatial computing R&D

**Talent**

Educational Curriculum

Incorporate spatial computing across **K-12 and higher education** with interdisciplinary focus

Training Programs

Launch **accelerated certifications** with industry leaders for rapid workforce development

Global Talent Attraction

Create **specialized visa pathways** for spatial computing professionals and entrepreneurs

Strategic Implementation Roadmap

2025-2026

2026-2027

2027-2028

2028-2029

2030

Foundation Phase

Establish regulations, initial infrastructure, and educational...

Acceleration Phase

Deploy digital twin, expand talent, launch

Growth Phase

Scale private sector adoption and international...

Integration Phase

Mainstream across industries and public services

Leadership Phase

Dubai established as global Touchable AI leader

Spatial Computing: The Next Technological Frontier

\$300B+

GLOBAL MARKET BY 2030

35%

ANNUAL GROWTH RATE

\$4B

DUBAI ECONOMIC IMPACT

- Market Opportunity**

Spatial computing will reach **\$300+ billion by 2030**, growing at 35% CAGR. Driven by AR/VR hardware advances, AI integration, and adoption across healthcare, manufacturing, retail, and education sectors.
- Dubai's Strategic Position**

Dubai is uniquely positioned as a **leading hub for spatial computing** with its Metaverse Strategy, forward-thinking regulations, world-class infrastructure, and commitment to technological innovation.
- Touchable AI: A Key Differentiator**

Touchable AI represents the next evolution in computing, where artificial intelligence becomes physically interactive and integrated into our environment rather than confined to screens, creating intuitive, spatial user experiences.

Spatial computing represents a **fundamental shift** in human-computer interaction. As digital and physical worlds merge, organizations that embrace this technology early will gain significant competitive advantages and define the future of innovation.

A Word from
Futurists Hub



Rashed Almazrooei

Co-founder, Futurists Hub

This research represents a pivotal moment in understanding how spatial computing will transform our region. Dubai's unique position at the intersection of technology, innovation, and global commerce presents an extraordinary opportunity to lead this revolution.

At Futurists Hub, we're committed to pioneering research that illuminates technological frontiers and guides strategic decision-making. Spatial computing is more than just another technology trend—it's a fundamental shift in how we'll interact with digital information and with each other. The organizations and regions that embrace this transformation early will define the next era of human-computer interaction and create tremendous value in the process.



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